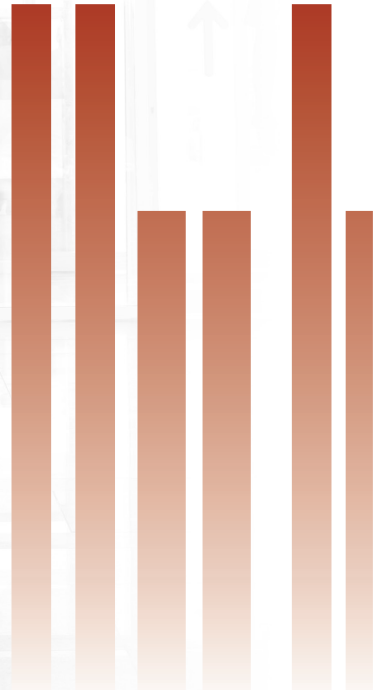


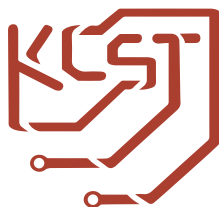


KUWAIT COLLEGE OF SCIENCE & TECHNOLOGY
كلية الكويت للعلوم والتكنولوجيا

Course Catalogue and Graduation Requirements

2021
— to —
2023





KUWAIT COLLEGE OF SCIENCE & TECHNOLOGY
كلية الكويت للعلوم والتكنولوجيا

Title	Course Catalogue and Graduation Requirements
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Course Catalogue and Graduation Requirement manual shall serve as the reference document for all students, academics, the registrar's office and departments directly or indirectly involved with students. The Quality Assurance and Accreditation Office shall retain this manual and be responsible for all updates and reviews.

The University reserves the right to make changes to the manual. Changes will be included in the addendum documentation and incorporated into the manual.

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Institutional Information

Our Vision

Kuwait College of Science and Technology's (KCST's) vision is to be among the leading universities of science and technology in the Middle East through the pursuit of academic excellence guided by human, social and environmental values.

Our Mission

Kuwait College of Science and Technology (KCST) is a private university established to provide internationally recognised undergraduate and graduate degree programs in science, technology and supporting disciplines. In fulfilling its mission, KCST offers state-of-the-art education and prepares students to be leaders in the use of multidisciplinary approaches to solve complex problems. This means that alongside the rigorous curricula in science and technology, containing significant amounts of hands-on experience and opportunities to take part in high-level research projects, courses in management, entrepreneurship and socially responsible leadership are also incorporated into the curricula to ensure a well-rounded exposure to the real-world of practice. In serving as a platform to create and propagate knowledge, nurture free and open inquiry and advance modern scientific thinking, KCST's mission includes the development of lasting bonds amongst students, faculty and the community.

Our Values

Professionalism: utilising knowledge and specialised training to make sound judgments and building environments and standards distinctive to the specialisation.

Quality: ensuring appropriate and effective teaching, supporting assessment and learning opportunities to students, directed by comprehensive procedures and stringent quality control

Excellence: promoting a robust and progressive strategic governance and management leading to high academic achievement standards resulting in exceptional student experience, positive stakeholder satisfaction and commitment to research and academic development.

Respect: providing an environment in which all members of the university community treat each other with dignity and respect and where bullying, harassment and discrimination are unacceptable.

Fairness: treating all and each other impartially, without judgement, prejudice or bias and creating an environment that allows each and everyone to do their very best work

History of KCST

2008

The College is licensed under the Amiri Decree No. 106 of 2008 by the Ministry of Higher Education (MOHE) as Kuwait Institute of Science and Technology.

2011-2015

KCST campus construction at Doha District, Block 7, Kuwait.

2015

Name changed to Kuwait College of Science and Technology by the Amiri Decree No 148 of 2015 on May 2015 20.

October 4, the first KCST President takes office.

October 21, PUC approval to start operations in Doha Campus.

November 20, PUC approval to open admission to students.

2016

January 17, classes started with the first 16 students in the Foundation program.

September 15, Degree programs commence. KCST is fully operational, with a total of 340 students enrolled in the degree and foundation programs.

Curricular Partner University

Indian Institute of Technology Delhi (IIT Delhi)

Indian Institute of Technology Delhi (IIT Delhi) was established in 1961 as a Centre of Excellence for training, research and development in science, engineering and technology. IIT Delhi was selected as an Institute of Eminence and is ranked 185 globally in the QS world university ranking and 54 in the QS subject ranking (Engineering and Technology).

The agreement between IIT Delhi and KCST started in 2004 to create the curriculum and all essential policies and procedures to secure approvals from the PUC for three programs. This part of the agreement was achieved in 2008. This was followed by Phase Two (2010-2016) of the agreement to support the setup of the College, including consultancy and advice on the building and furniture, specifications of laboratory and library space and recruitment of senior management positions, including the President and Dean of Faculty.

Phase Three of the agreement (2016-2020) focused on supporting teaching at the College and faculty recruitment. This period has seen adaptations and improvements to the curriculum to reflect advancements in the topics covered by the curriculum and the international standards. The period has seen an intensive level of interaction between the two institutions. The IIT Delhi team members reviewed the delivery of the curriculum. They paid several visits to the KCST campus to ensure the quality of the academic and functional operations of the College. The agreement has been extended to 2020-2025. The agreement covers quality monitoring and support for the expansion programs of the College and its plans for changing status to a full-scale university.

Accreditation Status

Institutional Accreditation

The PUC granted KCST Institutional Accreditation for five years (2021-2026). KCST is proud to be the first private university to receive Grade A accreditation in its first round, indicating its academic excellence and technological development and the great value this achievement adds to the work field.

ABET Accreditation

KCST has successfully received Accreditation Course Catalogue and Graduation Requirements for its **Engineering Programs**: Bachelor of Engineering in Computer Engineering and Bachelor of Engineering in Electronics and Communications Engineering until September 30, 2028.

* The accreditation action for these two newly accredited programs extends retroactively from October 1, 2019.

KSE Approval

Kuwait Society of Engineering (KSE) approved the applications of KCST graduates for KSE membership based on the ABET Accreditation.

Program Accreditation by the Ministry of Higher Education and Scientific Research, Jordan

The Ministry of Higher Education and Scientific Research, Jordan, issued the Recognition and Certificates Equivalency Directorate to Kuwait College of Science and Technology on July 5 2021.

Collaborations

Curriculum Development and Quality Monitoring

- Indian Institute of Technology Delhi (IIT D) – for engineering programs
- Bayes Business School (formerly CASS), City, University of London.

Research Collaboration and Student Exchange Agreements

- Kanagawa Institute of Technology, Japan
- Edith Cowan University, Australia
- California State University

Technology partners

- Huawei ICT Academy
- CISCO Networking Academy
- MIT FabLab
- EC-Council
- Alpha Solar Technologies, Australia



Degree Programs



Electronics and Communications Engineering Program



Electronics & Communications Engineering Program

The Electronics and Communications Engineering Program offers one undergraduate degree program: Bachelor of Engineering in Electronics & Communications Engineering.

Bachelor of Electronics and Communications Engineering– BEng ECE

The Electronics and Communications Engineering Program extends over four and a half years offered exclusively on a daytime, on-campus basis. The program is offered in ten semesters. In the summer semester of the third year, students are highly encouraged to participate in a practical training program with a local, regional, or international organization.

Program Mission

The mission of the undergraduate bachelor of Engineering program in Electronics and Communications Engineering is to deliver high-quality undergraduate education which combines balanced theoretical and practical topics in Electronics and Communications Engineering to provide students with the skills and abilities to pursue successful careers and grow into lifelong learners with the ability to provide professional services of the highest quality to the community, commit to the ethical and professional responsibilities, and expand, through teaching and research, the knowledge and technology bases in Electronics and Communications Engineering required at the global level while focusing on the needs of Kuwait and the region. The program shall also prepare students to serve the community in lead teaching, public and private sectors position.

Program Educational Objective

The Electronics and Communications Engineering program is designed to prepare students for continuous learning and fruitful professions.

Our graduates will be able to:

- Apply the knowledge and skills to perform at an engineer level and excel in a field related to their program.
- Pursue specialized training and research in their field and familiarize themselves with changing technology in the workplace.
- Engage in life-long learning, career growth, leadership, and community service work.
- Perform satisfactorily if they choose to continue their education in a master's program.
- Conduct themselves as ethical professionals with integrity, uphold social responsibility and promote sustainability.

The Student Outcomes (SOs)

The Student Outcomes (SOs) adopted by KCST for the ECE program are the EAC of ABET Student Outcomes.

Graduates of the program will have:

1. An ability to identify, formulate and solve complex engineering problems by applying engineering, science, and mathematics principles.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate with a range of audiences.
4. An ability to recognize ethical and professional responsibilities in engineering solutions and make informed judgements which must consider the impact of engineering solutions in global economic, environmental, and social contexts.
5. An ability to function effectively as a team member provides leadership, creates a collaborative and inclusive environment, establishes goals, plans tasks, and meets objectives.
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data and use engineering judgement to draw conclusions.
7. An ability to acquire and apply new knowledge as needed using appropriate learning strategies.

Electronics & Communications Engineering Graduation Requirement Summary

To earn the degree, students must complete the 144 credits in the prescribed time. The following table shows a summary of the graduation requirement and associated major categories:

Category	Core Credits	Elective Credits	Total number of Credits
Basic Sciences and Mathematics (BI/CH/PH/MA)	26	6	32
Languages, Social Sciences and Business Studies (LA/SS/BM)	18	12	30
Electronics and Communications Engineering (EE/CE/CS)	73	9	82
Total	117	27	144

Bachelor of Engineering in Electronics and Communications Engineering

Basic Sciences and Mathematics - Core and Electives

CORE REQUIREMENTS

Student must complete 26 credit hours from the following core courses

Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
MA101	Mathematics I	3	(3,0)	None
MA102	Mathematics II	3	(3,0)	MA101
MA203	Linear Algebra	3	(3,0)	None
MA204	Engineering Mathematics	3	(3,0)	MA102
MA301	Probability and Statistics	3	(3,0)	MA102
MA302	Ordinary Differential Equations	3	(3,0)	MA102
PH101	Physics I	4	(3,2)	None
PH102	Physics II	4	(3,2)	PH101

ELECTIVES REQUIREMENTS

Student must complete 6 credit hours from the following elective courses

Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
BI201	Introduction to Modern Biology	3	(3,0)	None
CH101	Physical Chemistry	3	(2,2)	None
MA111	Discrete Mathematics	3	(3,0)	None
MA213	Computational Linear Algebra	3	(2,2)	MA203
MA315	Operations Research	3	(3,0)	MA301
MA316	Complex Variables	3	(3,0)	MA203,MA204
MA318	Numerical Methods for Engineers	3	(2,2)	MA203,MA204
MA340	Abstract Algebra	3	(3,0)	MA203
MA350	Topology and Functional Analysis	3	(3,0)	MA203
PH103	Solid State Physics	3	(3,0)	PH102
PH201	Modern Physics for Engineers	3	(3,0)	PH102,MA102
PH301	Solid State Physics and Devices	3	(3,0)	PH103,PH201

Languages, Social Sciences and Business Studies - Core and Electives

CORE REQUIREMENTS

Student must complete 18 credit hours from the following core courses

Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
BM111	Introduction to Management	3	(3,0)	None
BM227	Leadership Competencies for Engineers	3	(3,0)	BM111
BM301	Innovation and Entrepreneurship	3	(3,0)	BM111
LA101	English I	3	(3,0)	None
LA103	English II	3	(3,0)	LA101
SS111	Introduction to Economics	3	(3,0)	None

ELECTIVES REQUIREMENTS

Student must complete 12 credit hours from the following elective courses

Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
BM201	Financial and Managerial Accounting	3	(3,0)	MA101
BM223	Managing Human Resources	3	(3,0)	BM111
BM228	Strategic Management	3	(3,0)	BM111
BM229	Project Management	3	(3,0)	BM111
BM230	Introduction to Industrial Management	3	(3,0)	BM111
BM421	Business Marketing and Sales	3	(3,0)	BM111, BM227
LA102	Arabic I	3	(3,0)	None
LA201	English for Engineers	3	(3,0)	LA101
SS101	History of Arab Civilization	3	(3,0)	None
SS104	History of Kuwait	3	(3,0)	None
SS212	Macroeconomics	3	(3,0)	SS111
SS221	Introduction to Psychology	3	(3,0)	None
SS231	Philosophy and Ethics	3	(3,0)	None
SS332	Philosophy of Science	3	(3,0)	None

Electronics and Communications Engineering - Core and Electives

CORE REQUIREMENTS				
Student must complete 73 credit hours from the following core courses				
Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
CE100	Introduction to Computer Systems	3	(2,2)	None
CE212	Digital Logic Design	4	(3,2)	CS101
CE222	Computer Architecture	4	(3,2)	CE212
CS101	Introduction to Computers and Programming	4	(3,2)	CE100 or CS100
CS112	Data Abstraction & Object-Oriented Programming	4	(3,2)	CS101
EE101	Introduction to Electrical Engineering	4	(3,2)	None
EE112	Electronic Instrumentation	3	(2,2)	EE101
EE201	Signals and Systems	4	(3,2)	MA102
EE211	Circuit Theory	4	(3,2)	EE101
EE212	Digital Electronics	4	(3,2)	EE101
EE312	Electronic Devices and Systems	4	(3,2)	EE212
EE341	Communication Systems	4	(3,2)	EE201
EE342	Data Communications	4	(3,2)	EE201,MA301
EE351	Control Systems and Instrumentation	4	(3,2)	EE201
EE352	Digital Communications	4	(3,2)	EE341
EE365	Power Electronics	3	(2,2)	EE112,EE312
EE491	Major Project, Part 1	4	(0,8)	Min** 90 Credit Hours, Co-req***: {GE480, and (EE365, or EE352)}
EE492	Major Project, Part 2	6	(0,12)	EE491
GE480	Ethical, Legal and Professional Issues for Engineers	2	(2,0)	Min** 60 Credit Hours

ELECTIVES REQUIREMENTS				
Student must complete 9 credit hours from the following elective courses				
Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
CE334	Computer Network Protocols and Applications	3	(3,0)	CE222
CE376	Mobile Applications Development	3	(2,2)	CS203
CE420	Advanced Computer Architecture	3	(3,0)	CE222
CE421	Embedded Systems	3	(2,2)	CE222
CS193	Software Tools	3	(2,2)	None
CS201	Critical Thinking and Problem Solving	2	(2,0)	None
CS301	Numerical and Scientific Computing	3	(2,2)	CS203, MA102
CS333	Computer Network Systems & Protocols	3	(3,0)	CE222
CS391	Software Engineering	3	(2,2)	CS371
EE360	Introduction to Satellite	3	(3,0)	EE201
EE362	Electromagnetic Waves and Theory	3	(2,2)	PH102,MA204
EE363	Analog and mixed Analog Integrated Circuit Design	3	(2,2)	EE312
EE401	Self- study	2	(0,4)	None
EE402	Minor Project	3	(0,6)	None
EE403	Minor Project	2	(0,4)	None
EE404	Minor Project	1	(0,2)	None
EE410	Digital Signal Processing	3	(2,2)	EE201
EE420	VLSI Design, Tools and Technology	3	(2,2)	EE212
EE441	Wireless Communications	3	(3,0)	EE352
EE442	Optical Communications	3	(3,0)	EE341
EE443	Wireless Networks and Protocols	3	(3,0)	EE341
EE444	Information Theory	3	(3,0)	EE341
EE451	Image Processing	3	(2,2)	EE410
EE452	Mechatronics	3	(2,2)	EE312,EE351
EE453	Robotics and Automation	3	(2,2)	EE351
EE467	RF Microelectronics	3	(2,2)	EE363
EE470	Advanced Photovoltaics	3	(2,2)	EE365
EE481	Medical Electronics	3	(3,0)	EE312,EE 201

(L,H)*:(Lecture, Hands-on)

Min** :Min number of credit hours required to be completed in order to register a specific course

Co-req***: Concurrent prerequisite

BEng. Electronics and Communications Engineering Sample Study Plan per Semester

BEng. Electronics and Communications Engineering - Semester 1 - Freshman

Course Code	Course name	Credit Hours	Prerequisites
CE100	Introduction to Computer Systems	3	None
EE101	Introduction to Electrical Engineering	4	None
LA101	English I	3	None
MA101	Mathematics I	3	None
PH101	Physics I	4	None
	Total credit hours	17	

BEng. Electronics and Communications Engineering - Semester 2 - Freshman

CS101	Introduction to Computers & Programming	4	CE100
LA103	English II	3	LA101
MA102	Mathematics II	3	MA101
PH102	Physics II	4	PH101
EE112	Electronic Instrumentation	3	EE101
	Total credit hours	17	

BEng. Electronics and Communications Engineering - Semester 3 (Summer Semester) **** - Freshman

Elective	Languages, Social Sciences and Business Studies	3	as required
Elective	Languages, Social Sciences and Business Studies	3	as required
	Total credit hours	6	

BEng. Electronics and Communications Engineering - Semester 4 - Sophomore

EE211	Circuit Theory	4	EE101
CE212	Digital Logic Design	4	CS101
MA203	Linear Algebra	3	None
MA204	Engineering Mathematics	3	MA102
SS111	Introduction to Economics	3	None
	Total credit hours	17	

BEng. Electronics and Communications Engineering - Semester 5 - Sophomore

EE201	Signals and Systems	4	MA102
EE212	Digital Electronics	4	EE201
CE222	Computer Architecture	4	CE212
MA301	Probability and Statistics	3	MA102
BM111	Introduction to Management	3	None
	Total credit hours	18	

BEng. Electronics and Communications Engineering - Semester 6
(Summer Semester)*** - Sophomore

BM227	Leadership Competencies for Engineers	3	BM111
	Total credit hours	3	

BEng. Electronics and Communications Engineering - Semester 7 - Junior

EE312	Electronic Devices and Systems	4	EE212
EE341	Communication Systems	4	EE201
EE351	Control Systems and Instrumentation	4	EE201
GE480	Ethical, Legal and Professional Issues for Engineers	2	Min** 60 Credit Hours
MA302	Ordinary Differential Equations	3	MA102
	Total credit hours	17	

BEng. Electronics and Communications Engineering - Semester 8 - Junior

EE352	Digital Communications	4	EE341
EE365	Power Electronics	3	EE112, EE312
EE342	Data Communications	4	EE201, MA301
CS112	Data Abstraction and Object Oriented Programming	4	CS101
Elective	Electronics and Communications Engineering	3	as required
	Total credit hours	18	

BEng. Electronics and Communications Engineering - Semester 9 - Senior

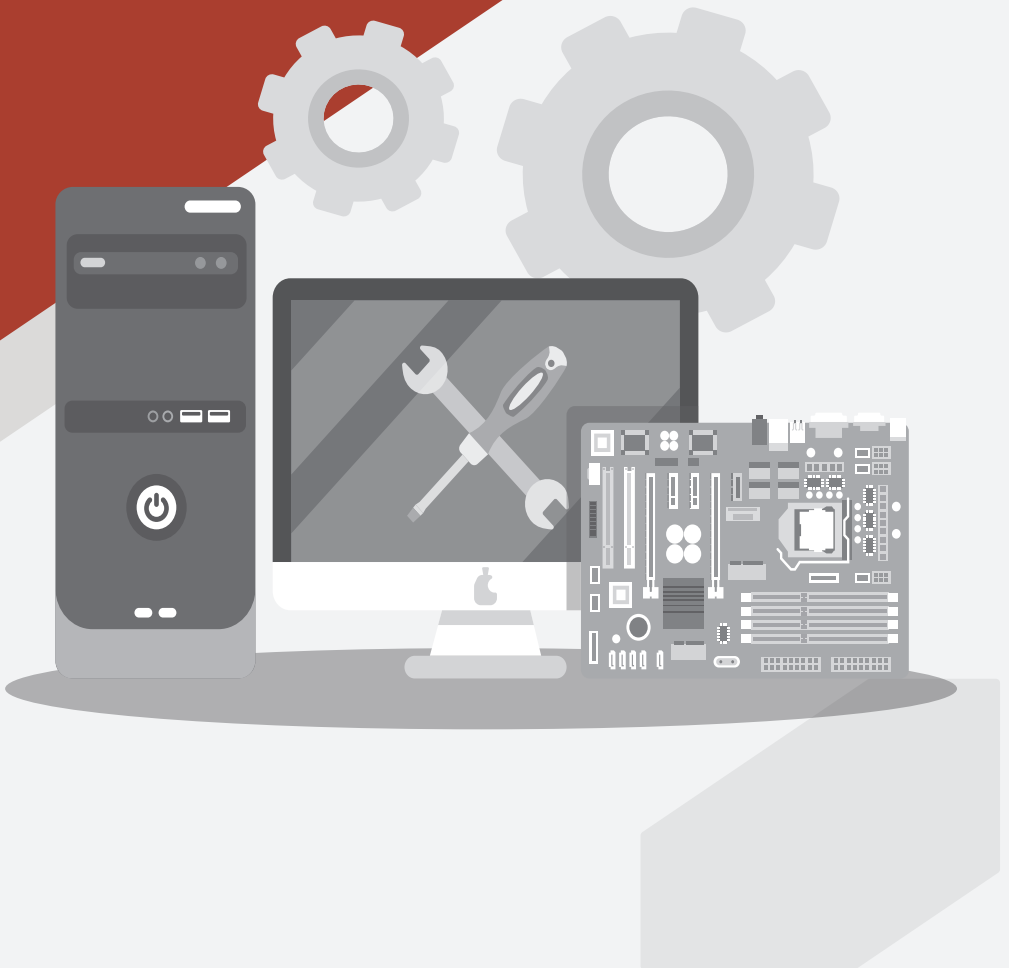
EE 491	Major Project Part 1	4	Min** 90 Credit Hours, Co-req***: {GE480, and (EE365, or EE352)}
BM301	Innovation and Entrepreneurship	3	BM111
Elective	Electronics and Communications Engineering	3	as required
Elective	Languages, Social Sciences and Business Studies	3	as required
Elective	Basic Sciences and Mathematics	3	as required
	Total credit hours	16	

BEng. Electronics and Communications Engineering - Semester 10 - Senior

EE492	Major Project Part 2	6	EE 491
Elective	Electronics and Communications Engineering	3	as required
Elective	Basic Sciences and Mathematics	3	as required
Elective	Languages, Social Sciences and Business Studies	3	as required
	Total credit hours	15	

(Summer Semester)****: optional semester for students willing to graduate in four academic years

Computer Engineering Program



Computer Engineering Program

The Computer Engineering Program offers one undergraduate degree program: Bachelor of Engineering in Computer Engineering.

Bachelor of Computer Engineering– BEng CE

The Computer Engineering Program extends over four and a half years offered exclusively on a daytime, on-campus basis. The program is offered in ten semesters. In the summer semester of the third year, students are highly encouraged to participate in a practical training program with a local, regional, or international organization.

Program Mission

The undergraduate Bachelor of Engineering program in Computer Engineering provides students with the skills and abilities to pursue successful careers and grow into lifelong learners with the ability to provide professional services of the highest quality to the community, commit to ethical and professional responsibilities, and expand, through teaching and research, the knowledge and technology base in Computer Science required at the global level while focusing on the needs of Kuwait and the region.

Program Educational Objective

The Computer Engineering program is designed to prepare students for continuous learning and fruitful professions.

- Graduates will have the knowledge and skills to perform at an engineer level and be employable in a field related to their program.
- Graduates of the Computer Engineering Program will be prepared for pursuing advanced study and research in their field, learning new skills, and familiarizing themselves with changing technology in the workplace.
- Graduates will engage in life-long learning, career growth, leadership, and community service to adapt to an ever-changing environment in Computer Engineering.
- Graduates choosing to pursue their education Masters' program will perform satisfactory in such programs.
- Graduates will conduct themselves ethically and with integrity, upholding social responsibility and promoting sustainability.

The Student Outcomes (SOs)

The Student Outcomes (SOs) adopted by KCST for the CE program are the EAC of ABET Student Outcomes.

Graduates of the program will have:

1. An ability to identify, formulate and solve complex engineering problems by applying engineering, science, and mathematics principles.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate with a range of audiences.
4. An ability to recognize ethical and professional responsibilities in engineering solutions and make informed judgements which must consider the impact of engineering solutions in global economic, environmental, and social contexts.
5. An ability to function effectively as a team member provides leadership, creates a collaborative and inclusive environment, establishes goals, plans tasks and meets objectives.
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data and use engineering judgement to draw conclusions.
7. An ability to acquire and apply new knowledge as needed using appropriate learning strategies.

Computer Engineering Graduation Requirement Summary

To earn the degree, students must complete the 144 credits in the prescribed time. The following table shows a summary of the graduation requirement and associated major categories:

Category	Core Credits	Elective Credits	Total number of Credits
Basic Sciences and Mathematics (BI/CH/PH/MA)	29	6	35
Languages, Social Sciences and Business Studies (LA/SS/BM)	18	12	30
Electronics and Communications Engineering (EE/CE/CS)	70	9	79
Total	117	27	144

Bachelor of Engineering in Computer Engineering

Basic Sciences and Mathematics - Core and Electives

CORE REQUIREMENTS

Student must complete 29 credit hours from the following core courses

Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
MA101	Mathematics I	3	(3,0)	None
MA102	Mathematics II	3	(3,0)	MA101
MA111	Discrete Mathematics	3	(3,0)	None
MA203	Linear Algebra	3	(3,0)	None
MA204	Engineering Mathematics	3	(3,0)	MA102
MA301	Probability and Statistics	3	(3,0)	MA102
MA302	Ordinary Differential Equations	3	(3,0)	MA102
PH101	Physics I	4	(3,2)	None
PH102	Physics II	4	(3,2)	PH101

ELECTIVES REQUIREMENTS

Student must complete 6 credit hours from the following elective courses

Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
BI201	Introduction to Modern Biology	3	(3,0)	None
CH101	Physical Chemistry	3	(2,2)	None
MA213	Computational Linear Algebra	3	(2,2)	MA203
MA315	Operations Research	3	(3,0)	MA301
MA316	Complex Variables	3	(3,0)	MA203,MA204
MA317	Number Theory	3	(3,0)	MA111
MA318	Numerical Methods for Engineers	3	(2,2)	MA203,MA204
MA340	Abstract Algebra	3	(3,0)	MA203
MA350	Topology and Functional Analysis	3	(3,0)	MA203
PH103	Solid State Physics	3	(3,0)	PH102
PH201	Modern Physics for Engineers	3	(3,0)	PH102,MA102
PH301	Solid State Physics and Devices	3	(3,0)	PH103,PH201

Languages, Social Sciences and Business Studies - Core and Electives

CORE REQUIREMENTS

Students must complete 18 credit hours from the following Core courses

Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
BM111	Introduction to Management	3	(3,0)	None
BM227	Leadership Competencies for Engineers	3	(3,0)	BM111
BM301	Innovation and Entrepreneurship	3	(3,0)	BM111
LA101	English I	3	(3,0)	None
LA103	English II	3	(3,0)	LA101
SS111	Introduction to Economics	3	(3,0)	None

ELECTIVES REQUIREMENTS

Students must complete 12 credit hours from the following basket of electives

Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
BM201	Financial and Managerial Accounting	3	(3,0)	MA101
BM223	Managing Human Resources	3	(3,0)	BM111
BM228	Strategic Management	3	(3,0)	BM111
BM229	Project Management	3	(3,0)	BM111
BM230	Introduction to Industrial Management	3	(3,0)	BM111
BM421	Business Marketing and Sales	3	(3,0)	BM111, BM227
LA102	Arabic I	3	(3,0)	None
LA201	English for Engineers	3	(3,0)	LA101
SS101	History of Arab Civilization	3	(3,0)	None
SS104	History of Kuwait	3	(3,0)	None
SS212	Macroeconomics	3	(3,0)	SS111
SS221	Introduction to Psychology	3	(3,0)	None
SS231	Philosophy and Ethics	3	(3,0)	None
SS332	Philosophy of Science	3	(3,0)	None

Computer Engineering - Core and Electives

CORE REQUIREMENTS Students must complete 70 credit hours from the following Core courses				
Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
CE100	Introduction to Computer Systems	3	(2,2)	None
CE212	Digital Logic Design	4	(3,2)	CS101
CE222	Computer Architecture	4	(3,2)	CE212
CE421	Embedded Systems	3	(2,2)	CE222
CE491	Major Project, Part 1	4	(0,8)	Min** 90 Credit Hours, Co-req***: {GE480, and CS391}
CE492	Major Project, Part 2	6	(0,12)	CE491
CS101	Introduction to Computers and Programming	4	(3,2)	CE100
CS112	Data Abstraction & Object-Oriented Programming	4	(3,2)	CS101
CS203	Data Structures and Objects	4	(3,2)	CS101
CS322	Operating Systems	4	(3,2)	CE222
CS333	Computer Network Systems & Protocols	3	(3,0)	CE222
CS371	Databases	4	(3,2)	CS203, MA111
CS391	Software Engineering	3	(2,2)	CS371
EE101	Introduction to Electrical Engineering	4	(3,2)	None
EE201	Signals and Systems	4	(3,2)	MA102
EE342	Data Communications	4	(3,2)	EE201, MA301
EE410	Digital Signal Processing	3	(2,2)	EE201
EE451	Image Processing	3	(2,2)	EE410
GE480	Ethical, Legal and Professional Issues for Engineers	2	(2,0)	Min** 60 Credit Hours

ELECTIVES REQUIREMENTS

Students must complete 9 credit hours from the following basket of electives

Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
CE334	Computer Network Protocols and Applications	3	(3,0)	CE222
CE364	Human-Machine Interaction	3	(2,2)	CS203
CE376	Mobile Applications Development	3	(2,2)	CS203
CE401	Self- study	2	(0,4)	None
CE402	Minor Project	3	(0,6)	None
CE403	Minor Project	2	(0,4)	None
CE404	Minor Project	1	(0,2)	None
CE420	Advanced Computer Architecture	3	(3,0)	CE222
CS193	Software Tools	3	(2,2)	None
CS201	Critical Thinking and Problem Solving	2	(2,0)	None
CS301	Numerical and Scientific Computing	3	(2,2)	CS203, MA302
CS341	Programming Languages	3	(2,2)	CS203, MA111
CS351	Computer Graphics	3	(2,2)	CS203, MA111
CS352	Multimedia Systems	3	(2,2)	CS322, EE201
CS361	Artificial Intelligence	3	(2,2)	CS203, CE222
CS372	Web Information Systems	3	(2,2)	CS203, MA111
CS373	Analysis and Design of Algorithms	3	(2,2)	CS203, MA111
CS413	Theory of Computation	3	(3,0)	CS203, MA111
CS426	Distributed Computation	3	(3,0)	CS322
CS439	Information Security	3	(2,2)	CS203
EE212	Digital Electronics	4	(3,2)	EE101
EE341	Communication Systems	4	(3,2)	EE201
EE352	Digital Communications	4	(3,2)	EE341
EE420	VLSI Design, Tools and Technology	3	(2,2)	EE212
EE441	Wireless Communications	3	(3,0)	EE352
EE443	Wireless Networks and Protocols	3	(3,0)	EE341
EE444	Information Theory	3	(3,0)	EE341

(L,H)*:(Lecture, Hands-on)

Min** :Min number of credit hours required to be completed in order to register a specific course

Co-req***: Concurrent prerequisite

BEng. Computer Engineering Sample Study Plan per Semester

BEng. Computer Engineering - Semester 1 - Freshman

Course Code	Course name	Credit Hours	Prerequisites
CE100	Introduction to Computer Systems	3	None
EE101	Introduction to Electrical Engineering	4	None
LA101	English I	3	None
MA101	Mathematics I	3	None
PH101	Physics I	4	None
	Total credit hours	17	

BEng. Computer Engineering - Semester 2 - Freshman

CS101	Introduction to Computers & Programming	4	CE100
LA103	English II	3	LA101
MA102	Mathematics II	3	MA101
PH102	Physics II	4	PH101
SS111	Introduction to Economics	3	None
	Total credit hours	20	

BEng. Computer Engineering - Semester 3 (Summer Semester)**** - Freshman

Elective	Languages, Social Sciences and Business Studies	3	as required
Elective	Languages, Social Sciences and Business Studies	3	as required
	Total credit hours	6	

BEng. Computer Engineering - Semester 1 - Freshman

CS112	Data Abstraction and Object Oriented Programming	4	CS101
CE212	Digital Logic Design	4	CS101
MA203	Linear Algebra	3	None
MA204	Engineering Mathematics	3	MA102
MA111	Discrete Mathematics	3	None
	Total credit hours	17	

BEng. Computer Engineering - Semester 5 - Sophomore

CS203	Data Structures and Objects	4	CS101
CE222	Computer Architecture	4	CE212
MA301	Probability and Statistics	3	MA102
BM111	Introduction to Management	3	None
Elective	Basic Sciences and Mathematics	3	as required
	Total credit hours	17	

BEng. Computer Engineering - - Semester 6 (Summer Semester)**** - Freshman

BM 227	BM111	3	BM111
Elective	as required	3	as required
		6	

BEng. Computer Engineering - Semester 7 - Junior

CS371	Databases	4	CS203, MA111
CS322	Operating Systems	4	CE222
EE201	Signals and Systems	4	MA102
GE480	Ethical, Legal and Professional Issues for Engineers	2	Min 60 credit hours
MA302	Ordinary Differential Equations	3	MA102
	Total credit hours	17	

BEng. Computer Engineering - Semester 8 - Junior

CE421	Embedded Systems	3	CE222
CS391	Software Engineering	3	CS371
EE410	Digital Signal Processing	3	EE201
EE 342	Data Communications	4	EE201, MA301
Elective	Computer Engineering	3	as required
	Total credit hours	16	

BEng. Computer Engineering - Semester 9 - Senior

CE 491	Major Project Part 1	4	Min 90 credit hours, and Co-Req {GE480 and CS391}
CS333	Computer Network Systems & Protocols	3	CE222
EE451	Image Processing	3	EE410
Elective	Computer Engineering	3	as required
BM 301	Innovation and Entrepreneurship	3	BM111
	Total credit hours	16	

BEng. Computer Engineering - Semester 10 - Senior

CE 492	Major Project Part 2	6	CE 491
Elective	Computer Engineering	3	as required
Elective	Basic Sciences and Mathematics	3	as required
Elective	Languages, Social Sciences and Business Studies	3	as required
	Total credit hours	15	

*(Summer Semester)****:optional semester for students willing to graduate in four academic years*

Computer Science Program



Computer Science Program

The Computer Science Program offers one undergraduate degree program: Bachelor of Science in Computer Science.

Bachelor of Computer Science – BSc CS

The Computer Science Program extends over four years and is offered exclusively on a daytime, on-campus basis. The program is offered in eight semesters. In the summer semester of the third year, students are highly encouraged to participate in a practical training program with a local, regional, or international organization.

Program Mission

The undergraduate Bachelor of Science program in Computer Science is to deliver high-quality undergraduate education which combines balanced theoretical and practical topics in Computer Science to provide students with the skills and abilities to pursue successful careers and grow into lifelong learners with the ability to provide professional services of the highest quality to the community, commit to ethical and professional responsibilities, and expand, through teaching and research, the knowledge and technology base in Computer Science required at the global level while focusing on the needs of Kuwait and the region.

Program Educational Objectives

- Graduates will have the knowledge and skills to perform at a professional level and be employable in a field related to their program.
- Employers will be satisfied with the performance and education of program graduates and will be willing to employ future graduates.
- Graduates of the Computer Science Program will be prepared for increasing levels of responsibility, learning new skills, and adapting to changing technology in the workplace.
- Graduates choosing to continue their education Master's program will perform satisfactorily in such programs.
- Graduates commit to active participation in life-long learning to adapt to an ever-changing environment in Computer Science.

Student Outcomes

The Student Outcomes (SOs) adopted by KCST for the CS program are the CAC of ABET Student Outcomes.

Graduates of the program will have the ability to:

- 1. Analyze a complex computing problem and apply computing principles and other relevant disciplines to identify solutions.
- 2. Design, implement and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
- 3. Communicate effectively in a variety of professional contexts.
- 4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
- 5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
- 6. Apply computer science theory and software development fundamentals to produce computing-based solutions

Computer Science Graduation Requirement Summary

Students must complete 126 credits in the prescribed time to earn the degree. The following table shows a summary of the graduation requirement and associated major categories:

Category	Core Credits	Elective Credits	Total number of Credits
Basic Sciences and Mathematics (BI/CH/PH/MA)	23	6	29
Languages, Social Sciences and Business Studies (LA/SS/BM)	18	12	30
Electronics and Communications Engineering (EE/CE/CS)	58	9	67
Total	99	27	126

Bachelor of Science in Computer Science

Basic Sciences and Mathematics - Core and Electives

CORE REQUIREMENTS

Student must complete 23 credit hours from the following core courses

Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
MA101	Mathematics I	3	(3,0)	None
MA102	Mathematics II	3	(3,0)	MA101
MA111	Discrete Mathematics	3	(3,0)	None
MA203	Linear Algebra	3	(3,0)	None
MA301	Probability and Statistics	3	(3,0)	MA102
PH101	Physics I	4	(3,2)	None
PH102	Physics II	4	(3,2)	PH101

ELECTIVES REQUIREMENTS

Student must complete 6 credit hours from the following basket of electives

Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
BI201	Introduction to Modern Biology	3	(3,0)	None
CH101	Physical Chemistry	3	(2,2)	None
MA213	Computational Linear Algebra	3	(2,2)	MA203
MA315	Operations Research	3	(3,0)	MA301
MA317	Number Theory	3	(3,0)	MA111
MA340	Abstract Algebra	3	(3,0)	MA203
MA350	Topology and Functional Analysis	3	(3,0)	MA203
PH103	Solid State Physics	3	(3,0)	PH102
PH201	Modern Physics for Engineers	3	(3,0)	PH102, MA102
PH301	Solid State Physics and Devices	3	(3,0)	PH103, PH201

Languages, Social Sciences and Business Studies - Core and Electives

CORE REQUIREMENTS

Student must complete 18 credit hours from the following Core courses

Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
BM111	Introduction to Management	3	(3,0)	None
BM227	Leadership Competencies for Engineers	3	(3,0)	BM111
BM301	Innovation and Entrepreneurship	3	(3,0)	BM111
LA101	English I	3	(3,0)	None
LA103	English II	3	(3,0)	LA101
SS111	Introduction to Economics	3	(3,0)	None

ELECTIVES REQUIREMENTS

Student must complete 12 credit hours from the following basket of electives

Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
BM201	Financial and Managerial Accounting	3	(3,0)	MA101
BM223	Managing Human Resources	3	(3,0)	BM111
BM228	Strategic Management	3	(3,0)	BM111
BM229	Project Management	3	(3,0)	BM111
BM230	Introduction to Industrial Management	3	(3,0)	BM111
BM421	Business Marketing and Sales	3	(3,0)	BM111, BM227
LA102	Arabic I	3	(3,0)	None
LA201	English for Engineers	3	(3,0)	LA101
SS101	History of Arab Civilization	3	(3,0)	None
SS104	History of Kuwait	3	(3,0)	None
SS212	Macroeconomics	3	(3,0)	SS111
SS221	Introduction to Psychology	3	(3,0)	None
SS231	Philosophy and Ethics	3	(3,0)	None
SS332	Philosophy of Science	3	(3,0)	None

Computer Science - Core and Electives

CORE REQUIREMENTS				
Student must complete 58 credit hours from the following Core courses				
Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
CE100	Introduction to Computer Systems	3	(2,2)	None
CE212	Digital Logic Design	4	(3,2)	CS101
CE222	Computer Architecture	4	(3,2)	CE212
CS101	Introduction to Computers and Programming	4	(3,2)	CE100/CS100
CS112	Data Abstraction & Object-Oriented Programming	4	(3,2)	CS101
CS203	Data Structures and Objects	4	(3,2)	CS101
CS322	Operating Systems	4	(3,2)	CE222
CS333	Computer Network Systems & Protocols	3	(3,0)	CE222
CS341	Programming Languages	3	(2,2)	CS203, MA111
CS371	Databases	4	(3,2)	CS203, MA111
CS373	Analysis and Design of Algorithms	3	(2,2)	CS203, MA111
CS391	Software Engineering	3	(2,2)	CS371
CS413	Theory of Computation	3	(3,0)	CS203, MA111
CS491	Major Project, Part 1	4	(0,8)	Min** 90 credit hours, and Co-Req {GE480 and CS391}
CS492	Major Project, Part 2	6	(0,12)	CS491
GE480	Ethical, Legal and Professional Issues for Engineers	2	(2,0)	Min** 60 Credit Hours

ELECTIVES REQUIREMENTS

Student must complete 9 credit hours from the following basket of electives

Course Code	Course name	Credit Hours	Contact Hours (L,H)*	Prerequisites
CE364	Human-Machine Interaction	3	(2,2)	CS203
CE376	Mobile Applications Development	3	(2,2)	CS203
CE420	Advanced Computer Architecture	3	(3,0)	CE222
CE421	Embedded Systems	3	(2,2)	CE222
CS193	Software Tools	3	(2,2)	None
CS201	Critical Thinking and Problem Solving	2	(2,0)	None
CS221	Computer Organization	4	(3,2)	CE212
CS301	Numerical and Scientific Computing	3	(2,2)	MA302, CS203
CS304	Numerical Computation Methods	3	(2,2)	MA302, CS203
CS351	Computer Graphics	3	(2,2)	CS203, MA111
CS352	Multimedia Systems	3	(2,2)	CS322, EE201
CS361	Artificial Intelligence	3	(2,2)	CS203, CE222
CS372	Web Information Systems	3	(2,2)	CS203, MA111
CS401	Self- study	2	(0,4)	None
CS402	Minor Project	3	(0,6)	None
CS403	Minor Project	2	(0,4)	None
CS404	Minor Project	1	(0,2)	None
CS426	Distributed Computation	3	(3,0)	CS322
CS439	Information Security	3	(2,2)	CS203
EE101	Introduction to Electrical Engineering	4	(3,2)	None
EE201	Signals and Systems	4	(3,2)	MA102
EE212	Digital Electronics	4	(3,2)	EE101
EE342	Data Communications	4	(3,2)	EE201,MA301

(L,H):(Lecture, Hands-on)*

*Min** :Min number of credit hours required to be completed in order to register a specific course*

*Co-reg***: Concurrent prerequisite*

Bachelor of Engineering in Computer Science - Sample Study Plan per Semester

Course Code	Course name	Credit Hours	Prerequisites
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BSc. Computer Science - Semester 1 - Freshman

CE100	Introduction to Computer Systems	3	None
LA101	English I	3	None
MA101	Mathematics I	3	None
PH101	Physics I	4	None
Elective	Languages, Social Sciences and Business Studies	3	as required
Total credit hours		16	

BSc. Computer Science - Semester 2 - Freshman

CS101	Introduction to Computers & Programming	4	CE100
LA103	English II	3	LA101
MA102	Mathematics II	3	MA101
PH102	Physics II	4	PH101
SS111	Introduction to Economics	3	None
Total credit hours		17	

BSc. Computer Science - Semester 3 - Sophomore

CS112	Data Abstraction and Object Oriented Programming	4	CS101
CE212	Digital Logic Design	4	CS101
MA111	Discrete Mathematics	3	None
MA203	Linear Algebra	3	None
BM111	Introduction to Management	3	None
Total credit hours		17	

BSc. Computer Science - Semester 4 - Sophomore

CS203	Data Structures and Objects	4	CS101
CE222	Computer Architecture	4	CE212
MA301	Probability and Statistics	3	MA102
BM 227	Leadership Competencies for Engineers	3	BM111
Elective	Basic Sciences and Mathematics	3	as required
Total credit hours		16	

BSc. Computer Science - Semester 5 - Junior

CS371	Databases	4	CS203, MA111
CS373	Analysis and Design of Algorithms	3	CS203, MA111
CS322	Operating Systems	4	CE222
GE480	Ethical, Legal and Professional Issues for Engineers	2	Min 60 credit hours
Elective	Languages, Social Sciences and Business Studies	3	as required
Total credit hours		16	

BSc. Computer Science - Semester 6 - Junior

CS391	Software Engineering	3	CS371
CS341	Programming Languages	3	CS203, MA111
CS413	Theory of Computation	3	CS203, MA111
BM 301	Innovation and Entrepreneurship	3	BM111
Elective	Computer Science	3	as required
Total credit hours		15	

BEng. Computer Engineering - Semester 7 - Senior

CS 491	Final Year Project Part 1	4	Min 90 credit hours, and Co-Req {GE480 and CS391}
CS333	Computer Network Systems & Protocols	3	CE222
Elective	Languages, Social Sciences and Business Studies	3	as required
Elective	Computer Science	3	as required
Elective	Basic Sciences and Mathematics	3	as required
Total credit hours		16	

BSc. Computer Science - Semester 8 - Senior

CS492	Final Year Project Part 2	6	CS491
Elective	Computer Science	3	as required
Elective	Languages, Social Sciences and Business Studies	3	as required
Total credit hours		12	



Foundation Programs



Program Overview:

All applicants are required to take the KCST Entrance Test at the admission stage. If an applicant does not obtain the minimum required scores in the Entrance Test for direct entry into the degree program, they will need to attend Foundation Program at the level determined by the University.

The Foundation Program is offered at two levels, each with a duration of one semester. Each Foundation Program level has four courses: English, Math, Physics, and Introduction to IT. A student admitted to the Foundation Program is required to pass all the assigned courses with a satisfactory grade to be eligible to enroll in the BSc or BEng programs.

Program Mission:

The mission of the Foundation Program is to help students achieve a level of proficiency that is expected for starting the BSc or BEng programs. It allows students to catch up on material that may not have been included in their high school curriculum and reviews material in the best high school curricula. The Foundation program also allows students with weak English backgrounds to improve their language skills.

Program Educational Objectives:

The program aims to assist the students:

- achieve the level of expected proficiency for starting the BSc or BEng programs.
- attain an acceptable level of English writing and communication
- accomplish the required competence in Math and Physics

Sample Study Plan:

Each Foundation semester typically has a -16 week duration, consisting of 14 weeks of instruction and two weeks set aside for mid-term and final exams.

The Foundation Program is offered in Level One and Level Two, and their entrance exam marks determine the level a student is admitted in.

Level One	Level Two
IT 001	IT 002
MA 002	MA 003
PH 002	PH 003
LA 002	LA 003

FOUNDATION Level One			
Course code	Course name	Hours of instruction	Credits
LA 002	English 002	8 hours/week	0
MA 002	Math 002	3 hours/week	0
PH 002	Physics 002	3 hours/week	0
IT 001	Computer Skills	2 hours/week	0

FOUNDATION Level Two			
Course code	Course name	Hours of instruction	Credits
LA 003	English 003	8 hours/week	0
MA 003	Math 003	3 hours/week	0
PH 003	Physics 003	3 hours/week	0
IT 002	Introduction to IT	2 hours/week	0